Muhammad Moosa Raza

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RESEARCH PROFILE

Early-career researcher in geospatial environmental science specializing in remote sensing–driven climate analysis, carbon dynamics, and large-scale Earth observation modeling.

Experienced in multi-decadal satellite data processing, machine learning-based classification, and ecosystem carbon estimation, with a strong focus on building reproducible, scalable geospatial workflows using Google Earth Engine and Python.

Research goal: advancing data-driven environmental modelling integrated with physical climate processes.

CORE RESEARCH COMPETENCIES

- Multi-temporal remote sensing (Landsat, Sentinel-2)
- Machine learning for EO (Random Forest, SVM, CART)
- Carbon stock modelling & ecosystem services (InVEST)
- Large-scale geospatial processing (Google Earth Engine)
- Spatial analysis & land system dynamics
- Environmental data pipelines & reproducible workflows

EDUCATION

M.Sc. Ecology and Nature Management / Climate Project Management Peoples' Friendship University of Russia (RUDN University) | *Graduated: June 8, 2026*

Degree conferred with Honors-level performance - GPA: 84.68 / 100 (ECTS Grade B)

Thesis: *Improving Long-Term Land Use/Land Cover Change Detection and Carbon Stock Estimation in Lahore, Pakistan (Grade: Excellent)*

B.Sc. Environmental Engineering - University of Agriculture Faisalabad

Focus: Hydrology · GIS · Environmental systems · Water quality

RESEARCH EXPERIENCE

Master's Thesis Researcher Remote Sensing & Carbon Modelling RUDN University | *2024 – 2026*

- Designed multi-decadal (1993–2023) LULC analysis pipeline using Landsat imagery in Google Earth Engine
- Implemented machine learning classification (Random Forest) achieving Accuracy: 97.8%–99.2%, Kappa: 0.971–0.989
- Quantified urban expansion (+117%) and vegetation loss (-27,639 ha)
- Estimated ecosystem carbon dynamics: total loss of ~3.15 million metric tons
- Integrated InVEST model for validation
- Built automated, reproducible geospatial workflows using GEE

Key contribution: *Demonstrated how urban land transitions drive large-scale carbon loss in rapidly developing regions*

PUBLICATIONS

Raza, M. M. et al. Three Decades of Urban Expansion and Ecosystem Carbon Loss in Lahore District, Pakistan: A Random Forest and InVEST-Based Assessment (1993–2023).

Submitted to Pakistan Journal of Science - under review, not yet published

PROJECTS

LULC Change Detection & Carbon Stock Estimation Lahore District, Pakistan

Google Earth Engine pipeline for multi-year (1993–2023) LULC classification using Random Forest and carbon stock estimation aligned with InVEST Carbon Model logic; underlying data for the submitted manuscript above.
github.com/moosarazauaf/gee-lulc-pakistan

Layyah District Flood Monitor

Web-based flood monitoring tool combining a FastAPI backend and Earth Engine pipeline: Sentinel-1 SAR flood-extent detection against a permanent water mask, restricted to the river floodplain buffer, with a Leaflet map front end.
github.com/moosarazauaf/layyah-flood-monitor

PROJECT-LEVEL SKILLS

- Spatio-temporal environmental modelling
- Handling large EO datasets (cloud-based processing)
- Accuracy assessment (confusion matrix, Kappa statistics)
- Linking land-use change → carbon cycle impacts
- Translating geospatial outputs into policy-relevant insights

TECHNICAL SKILLS

- Geospatial: Google Earth Engine, QGIS
- Programming: Python (data analysis, raster processing)
- Visualization: Matplotlib, Excel, R
- Languages: English (Fluent), Russian (Basic)

SCIENTIFIC COMMUNICATION

Speaker - International Student Scientific Seminar | *2025*

Topic: Urban Anthropogenic Impacts on Carbon Cycle

Participant - WasteTech 2025 (Moscow) | *2025*

Focus: sustainability & environmental technologies

CERTIFICATIONS

- Python Programming — University of Michigan
- Environmental Management Systems (EMS)